

Strict Architectural Boundary & Baseline Scope: In strict alignment with the source document, UAT scope is limited exclusively to baseline API capabilities. Features categorized as future enhancements (Nutrition Tracking, BMI Calculation, Wearable Device Integration, Workout Reminders, and Fitness Analytics Dashboards) are explicitly excluded.

1. UAT GOVERNANCE, SCOPE & EXECUTION PARAMETERS

EXECUTION PURPOSE

Validate end-to-end functionality, data contract compliance, cryptographic isolation, and Google Gemini AI responses against documented baseline system requirements.

SINGLE USER ROLE

Authenticated User: Singular role utilizing Bearer JWT authorization to perform workout record management and query personalized AI generative fitness insights.

EXECUTION STACK

Runtime: Node.js (v16+)
Framework: Express.js
Database: MongoDB / Mongoose
API Clients: Postman / ThunderClient

Strict Execution Precedence Rule: Testing execution strictly requires progressive execution across documented layers: **1. Authentication APIs** → **2. Workout Management APIs** → **3. AI Integration APIs**.

2. TEST SUITE 1: AUTHENTICATION & USER PROFILE EXECUTION

Verification of user onboarding, bcrypt.js password hashing, JWT credential provisioning, and bearer authorization enforcement.

TEST ID	FEATURE	TEST SCENARIO	PRECONDITIONS	TEST STEPS	EXPECTED RESULT	ACTUAL	STATUS
UAT-AUTH-01	User Registration	Successful onboarding of a new user with valid fields.	MongoDB running; email is unique in database.	Send POST /api/auth/register with body: {name, email, password}.	HTTP 201 Created. User registered; password hashed via bcrypt.js; returns user profile and JWT.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-AUTH-02	User Registration	Registration rejection due to duplicate email address.	User record already exists with targeted email.	Send POST /api/auth/register using an existing email address.	HTTP 400 Bad Request. System prevents duplicate account and returns standardized error message.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-AUTH-03	User Authentication	Successful authentication and issuance of Bearer JWT.	User account registered and verified in MongoDB.	Send POST /api/auth/login with matching registered email and password.	HTTP 200 OK. Returns authenticated user payload along with signed JSON Web Token (JWT).	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-AUTH-04	User Authentication	Rejection of login request due to invalid credentials.	User record registered.	Send POST /api/auth/login with correct email but invalid password.	HTTP 400 / 401 Unauthorized. Access denied; returns explicit authentication failure message.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-AUTH-	Profile Fetch	Retrieval of authenticated	Valid JWT acquired via	Send GET /api/auth/profile	HTTP 200 OK. Returns	NOT SPECIFIED	NOT SPECIFIED

3. TEST SUITE 2: WORKOUT LIFECYCLE & SEARCH EXECUTION

Verification of Workout CRUD operations, strict user data isolation, schema validations, and multi-criteria query filtering.

TEST ID	FEATURE	TEST SCENARIO	PRECONDITIONS	TEST STEPS	EXPECTED RESULT	ACTUAL	STATUS
UAT-WRK-01	Create Workout	Persist new workout record with full valid schema fields.	User authenticated; valid Bearer JWT present.	Send POST /api/workouts with body: {name, category, duration, caloriesBurned, date}.	HTTP 201 Created. Workout persisted with foreign key reference to logged-in user user Id.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-WRK-02	Create Workout	Validation rejection on missing required fields.	User authenticated; valid Bearer JWT present.	Send POST /api/workouts omitting required duration or name.	HTTP 400 Bad Request. Mongoose validation triggers; operation rejected with descriptive validation error.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-WRK-03	Read All Workouts	Retrieve all logged workouts strictly for authenticated user.	User authenticated; multiple workouts exist across different users.	Send GET /api/workouts with Authorization header.	HTTP 200 OK. Returns array containing workouts strictly bound to authenticated user's ID.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-WRK-04	Read Workout by ID	Retrieve distinct single workout record by valid MongoDB ObjectId.	Target workout belongs to logged-in user.	Send GET /api/workouts/:id with valid target ObjectId and Bearer JWT.	HTTP 200 OK. Returns targeted workout JSON document matching the route ID parameter.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-WRK-05	Read Workout by ID	Cross-user access restriction and unauthorized record traversal.	Workout record belongs to User B; token belongs to User A.	Send GET /api/workouts/:id for User B's record using User A's JWT.	HTTP 404 Not Found or 403 Forbidden. Cross-tenant access denied; record concealed.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-WRK-06	Update Workout	Modify attributes of an existing workout entry.	Workout record belongs to logged-in user.	Send PUT /api/workouts/:id with updated duration and caloriesBurned.	HTTP 200 OK. Targeted document updated in MongoDB; returns modified workout object.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-WRK-07	Delete Workout	Purge existing workout record from system.	Workout record belongs to logged-in user.	Send DELETE /api/workouts/:id with Bearer token.	HTTP 200 OK. Document permanently deleted; returns confirmation message payload.	NOT SPECIFIED IN SOURCE DOC	NOT VERTICAL

4. TEST SUITE 3: GOOGLE GEMINI AI INTELLIGENCE SERVICES

Verification of AI recommendation workflows, contextual fitness prompt construction, error resilience, and insights generation.

TEST ID	FEATURE	TEST SCENARIO	PRECONDITIONS	TEST STEPS	EXPECTED RESULT	ACTUAL	STATUS
UAT-AI-01	AI Workout Recommendation	Generate personalized workout regimen based on user goals.	Authenticated user; valid GEMINI_API_KEY configured.	Send POST /api/ai/recommend with body: {goal, fitnessLevel}.	HTTP 200 OK. Returns generative workout recommendation payload synthesized by Google Gemini API.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-AI-02	AI Fitness Insights	Derive intelligent performance insights from user's workout history.	Authenticated user has historical workouts logged in MongoDB.	Send GET /api/ai/insights with valid Bearer JWT.	HTTP 200 OK. Controller aggregates user workouts and returns Gemini-generated analysis and trends.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED
UAT-AI-03	AI Resiliency / Error Handling	Graceful fallback handling when external AI service fails or key is missing.	Invalid API key configured or network interruption simulated.	Send POST /api/ai/recommend with standard payload.	HTTP 500 Internal Server Error handled gracefully by centralized error middleware; returns sanitized message.	NOT SPECIFIED IN SOURCE DOC	NOT SPECIFIED

5. FORMAL UAT EXECUTION SIGN-OFF & APPROVALS

Sign-off confirms that all functional verification activities conform strictly to documented specifications and architectural requirements.

Stakeholder Role	Representative Name	Execution Decision	Approval Date	Signature
Lead Backend Engineer	Not specified in the source document	PENDING EXECUTION	Not specified in source doc	_____
Quality Assurance Lead	Not specified in the source document	PENDING EXECUTION	Not specified in source doc	_____
Technical Product Owner	Not specified in the source document	PENDING EXECUTION	Not specified in source doc	_____

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